Static Tokens Are Colder: Averaged Noise for Frame-Shared Video Generation

Errata and a follow-up to "Entropy-Continuum Mechanics for Video"

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Abstract. A recent report, *Entropy-Continuum Mechanics for Video*, treats video information as a moving material and derives a permanence score and *frame-shared generation*, in which a video diffusion transformer computes static-background tokens once and copies them across frames. We audit the report and build on it. The audit lists 17 errors and gaps, most of them verified numerically. The score charges pure noise 0.531 bits per pixel that it states are free; it is unbounded and scales inversely with a clip's grain; its clip-level maximum grows with clip length; proportional attention is exact only without temporal position information; and the headline fidelity of frame sharing, 41.5–43.8 dB, is measured against a reference the method has itself altered, while against the original output the report's own figure is 11–15 dB, below every baseline. Our new work answers the report's open question: is a sample generated from shared noise as good? Not when the noise is shared by copying. The F copies of a static token are F measurements of one quantity, the network pools them, and one measurement counted F times makes it over-confident by $\sqrt{F}$. We prove that for an ideal denoiser, replacing the copies by their frame *mean* leaves the generated sample exactly unchanged, at the start or mid-generation, whereas copying inflates the variance of Gaussian static content by exactly F. Both hold to float precision in a solvable model; with an exact nonlinear denoiser, copying returns the original clip in 0.1 % of runs and averaging in 90.2–100.0 %. In a trained 1.3M-parameter transformer, copying over-disperses backgrounds 2.6× and lands 7.9 dB from the original sample. Averaging removes the over-dispersion and gains 13 dB, but a network that has never seen tied tokens over-produces texture, and on equal footing frame sharing then loses to simply taking fewer steps. A short fine-tune with a regulariser that encodes the sufficiency of the frame mean closes the gap: with the base report's own 2-step draft map, mean-tied sharing reproduces the original sample at 43.9 dB with 2.0× fewer FLOPs and at 39.9 dB with 2.7× (oracle map: 43.8 dB, 3.2×), more than 15 dB above the fewer-steps baseline, while copying stays at 7.3 dB. The regulariser is not free: a no-regulariser control shows it, not extra training, causes the change, and it costs texture in the model's samples (0.74 to 0.54 of the data's level, against 0.85), a trade-off we chart but do not resolve. All experiments are toy-scale and CPU-only.

1 Introduction

A recent technical report, *Entropy-Continuum Mechanics for Video* [1] (the *base paper* below), treats the information in a video as a material that moves with the scene. It draws two tools from that picture. The first is a training-free *permanence violation score* (PVS) that looks for information created away from occlusion boundaries. The second is *frame-shared generation*: in a video diffusion transformer, the tokens of a static background are computed in one leader frame and copied to the others, with a log-multiplicity attention bias (proportional attention [2]) standing in for the removed tokens. The report is candid about its scale and its failures, and it ends on an open question: sharing forces the copies of a static token to start from the same noise, this "yields a different sample", and "whether the shared-noise sample is as good is untested".

This follow-up does two things.

Errata (§2). We list 17 errors, evaluation flaws and gaps in the base paper and verify most of them numerically. The most consequential one concerns the headline result of frame sharing. Its fidelity of 41.5–43.8 dB is measured against full compute from noise that the method has already modified, while every baseline it is compared with is measured against the original output. Against the original output the base paper's own number is 11–15 dB, below every baseline in its table.

An answer to the open question (§3–6). Copying one frame's noise to F frames is not a neutral way to make tokens identical. The F copies of a static token are F noisy measurements of one quantity, and the network has learned to pool them. The pooled measurement is their *mean*, whose noise is $\sqrt{F}$ times weaker than that of any single copy: static tokens are colder. Handing the network one copy while telling it, through proportional attention, that it holds F of them makes it over-confident by $\sqrt{F}$. We show:

- **Theory.** For an ideal denoiser and a static region, replacing the copies by their frame mean leaves the generated sample *exactly* unchanged, at the start or at any later step (Theorem 1). Replacing them by one copy runs the sampler with noise $\sqrt{F}$ too strong; for Gaussian content the output variance is inflated by exactly F (Theorem 2).

- **Network-free checks.** In an exactly solvable Gaussian model the law holds to three digits. With an exact nonlinear denoiser, copying returns the original clip in 0.1 % of runs and averaging in 90.2–100.0 %.

- **A trained transformer.** Copying over-disperses backgrounds 2.6× and lands 7.9 dB from the original sample; averaging removes the over-dispersion and reaches 20.9 dB.
- **What does not work.** On a network trained in the standard way, averaging is not sample-preserving either: tied tokens are outside its training distribution and it answers with 2.4× too much texture. On an equal footing, frame sharing then loses to simply taking fewer steps, with either way of tying. The base paper's central comparison does not survive.
- **What makes it work.** The ideal denoiser ignores the per-frame deviations; a trained one does not. A 3k-step fine-tune with a regulariser that penalises this dependence (Eq. 4) makes mean-tied sharing reproduce the original sample at 43.9 dB with 2.0× fewer FLOPs using the base paper's 2-step draft map, and at 43.8 dB with 3.2× using an oracle map; the fewer-steps baseline is at 27.3 dB and 23.6 dB for 2× and 2.9×. Copying stays at 7.3 dB. The regulariser is not free: the fine-tuned model's samples lose texture (Table 4).

The change to the base method is one line of code, and it leaves the compute saving untouched. All experiments are toy-scale and CPU-only, like those of the base paper; we report them as evidence about a mechanism.

2 Errata for the base paper

We read the base report [1] closely and re-derived or re-implemented what could be checked without the authors' code. Table 1 lists what we found; the script `experiments/e0_errata_checks.py` reproduces every number quoted in this section. We separate items we verified (analytically or numerically) from concerns that need the original code to quantify. None of this detracts from the report's candour: several items below sharpen negative results that it already states.

The score charges noise (item 1). For a Laplacian residual with scale b, $u = |r|/b$ is exponential with unit rate and Eq. 3 reads $s = \max(u-1,0)/\ln 2$. Before the clamp its mean is zero, which is the property the paper states. After the clamp it is $\int_1^\infty (u-1)e^{-u}du/\ln 2 = 1/(e\ln 2) = 0.531$ bits per pixel. At the 3.6–4.2 bits/px the paper quotes for its clips this alone implies $\mathrm{PVS}_t \approx 0.13$ for a clip in which nothing happens, twenty to forty times the 0.003–0.007 shown for legal clips in its Fig. 1. Our re-implementation of Eqs. 1, 3 and 4 gives 0.13 for an event-free noisy scene, as predicted. The printed equations therefore do not determine the printed numbers. The likely reason is that the synthetic clips are noise-free, so the residual is exactly zero on the static background; but then $b = \mathrm{median}|r|/\ln 2 = 0$ and Eq. 3 is undefined without a floor that the paper does not report.

The score is not a fraction, and depends on grain (items 3–4). Eq. 3 is linear in $|r|$ with slope $1/(b\ln 2)$ bits per grey level, however small b is. A code can always fall back to sending the pixel without reference to the previous frame, so a consistent "created information" is at most the intra cost ι of the new pixel. Without that cap, cleaner footage is charged more for the same change (Fig. 1c). This is a confound for the real-versus-generated test, where one class has camera grain and the other does not, and it matches the paper's remark that compression doubled the peak score for identical content.

The headline fidelity uses a moved reference (item 12). This is the most consequential item and the starting point of our own work. The † footnote of the base paper's Table 2 states the reference honestly, but the comparison drawn from it in Fig. 4 does not hold. The tied-noise reference also rewards over-sharing: marking more positions static pulls the reference and the output towards the same frozen video, so the metric cannot see the method's main failure mode. We reproduce both numbers in §6: copy-tied sharing agrees with its own tied reference to 61.2 dB and with the original sample to 7.9 dB.

3 Why copied noise is too hot

Setting. A clip is $x_0 \in \mathbb{R}^{F\times P\times D}$: F frames, P spatial positions, D values per token. Rectified flow corrupts it as $x_t = (1-t)\,x_0 + t\,\varepsilon$ with $\varepsilon \sim \mathcal{N}(0,I)$; in our convention $t=1$ is pure noise. The ideal denoiser is $D(x,t) = \mathbb{E}[x_0 \mid x_t = x]$, the probability-flow velocity is $u(x,t) = (x - D(x,t))/t$, and the K-step Euler sampler is $x^{i+1} = x^i - \frac{1}{K}\,u(x^i, t_i)$ with $t_i = 1 - i/K$ and $x^0 = \varepsilon$. We write $\Phi(\varepsilon) = x^K$ for its output.

Let S be a set of positions at which the clip does not change over time, $x_0[f,p] = b_p$ for every frame f and $p \in S$. For any x and $p \in S$ define the frame mean $\bar{x}_p = \frac{1}{F}\sum_f x[f,p]$ and the deviations $\delta_{f,p} = x[f,p] - \bar{x}_p$; x_R denotes all tokens outside S. Frame sharing needs the F copies of a static position to be identical, so the base paper overwrites them with the leader frame's noise. We compare two ways of making them identical:

$$\begin{aligned} \textsc{copy}: &\quad x[f,p] \leftarrow x[1,p], \\ \textsc{mean}: &\quad x[f,p] \leftarrow \bar{x}_p, \qquad\qquad p \in S. \end{aligned}$$

Lemma 1 (The frame mean is sufficient)**.** *If the data are static on S almost surely, then for every $t \in (0,1]$ the denoiser $D(x,t)$ depends on x only through $(\bar{x}_S, x_R)$, and its value at a position $p \in S$ is the same in every frame.*

Proof. For $p \in S$, $\sum_f \|x[f,p] - (1-t)b_p\|^2 = F\,\|\bar{x}_p - (1-t)b_p\|^2 + \sum_f \|\delta_{f,p}\|^2$. The likelihood of x given x_0 is therefore $\exp(-\|\delta\|^2/2t^2)$ times a function of $(\bar{x}_S, x_R, x_0)$. The first factor does not involve x_0 and cancels in the posterior. □

Lemma 2 (The sampler splits)**.** *Under the assumption of Lemma 1, along the Euler sampler (i) the pair $(\bar{x}_S, x_R)$ evolves by itself, and (ii) $\delta^i = t_i\,\delta^0$, so that $\delta^K = 0$.*

Proof. On S, $u[f,p] = (x[f,p] - \hat{b}_p)/t$ where $\hat{b}_p$ is the common denoiser output, a function of $(\bar{x}_S, x_R)$ by Lemma 1.

Table 1: Errata. Status: **A** shown analytically, **N** verified numerically here, **T** follows from the base paper's own text or tables, **C** concern that needs the authors' code to quantify.

	Where	Statement or practice	Finding	
Permanence score (PVS)				
1	Eq. 3	"Motion explained by noise alone costs nothing."	False. The clamp at zero leaves $\mathbb{E}[s] = 1/(e\ln 2) = 0.531$ bits/px under the paper's own Laplacian noise model ($2^{-\delta}$ times that with dead zone δ).	A, N
2	Eq. 3 vs. Eq. 2	Eq. 3 is called the Lagrangian form of the balance law.	It is not. The Lagrangian form of Eq. 2 concerns the scalar density, which the paper shows fails (0.57 AUC). Eq. 3 is a noise-normalised motion-compensated residual, i.e. warping error in other units; only the mask comes from the balance law.	A
3	Eq. 3–4	PVS is read as a share of the frame's information.	Eq. 3 charges a pixel without bound, so PVS is not a fraction (the paper itself reports 2.22). A legal soft spotlight scores 8.6 under the paper's σ=2 prefilter, with single pixels charged 91 bits; capping by the intra cost gives 0.33.	N, T
4	Eq. 3–4	One threshold for all footage.	Numerator is relative to the clip's noise, denominator is absolute bits: one fixed event scores 0.14, 0.46, 1.46 at noise $\sigma = 3, 1.5, 0.75$ grey levels. Clean (generated, denoised, blurred) footage is scored far more harshly than grainy footage.	N
5	§4, §4.3	Clip score is $\max_t \mathrm{PVS}_t$.	Grows with clip length. Two event-free sets with 127 and 15 frame pairs are separated with AUC $T_1/(T_1+T_2) = 0.89$. The synthetic and real clips differ in length.	A, N
6	§4.3	"There was no separation" (real vs. generated).	4 real and 15 generated clips, no violation labels, mixed lengths and noise levels (items 4, 5): the test can show neither separation nor its absence. §7 calls it "19 clips".	T
7	Table 1	AUC differences of 0.01–0.03 are interpreted.	No uncertainty is reported. With about 150 clips per class the standard error of an AUC near 0.8 is about 0.026.	A
8	§4.1–4.2	Headline 0.86 (clean) and 0.80 (stress).	As far as the text specifies, the two use different pipelines (prefilter off / on); no single configuration is reported on both sets.	C
9	§4.2	Occlusion boundaries from the neighbouring frame pairs.	Fixes one-frame events only. A melt, fade or boil spans the neighbouring pairs as well, so the spurious boundaries return.	C
10	§3–4	Sources are legal at occlusion boundaries.	*Any* creation there is free: an object that changes identity behind an occluder, or never re-emerges, costs nothing. The balance law has no storage term.	A
Frame-shared generation				
11	Eq. 5	Proportional attention "reproduces full attention exactly".	Only without temporal position information, which a video model needs. Random transformer: relative error 4.0×10^{-10} without a temporal embedding, 0.28 with one. The paper's own oracle result (43.6 dB, falling as more frames share) is not exact either.	N, T
12	Table 2, Fig. 4	Sharing "lies clearly above the fewer-steps baseline" (41.5–43.8 dB vs. 27–30 dB).	Different references. Baselines are scored against the original output; sharing is scored against full compute from noise the method has already tied. Against the original, sharing scores 11–15 dB (§5 of the paper), below every baseline in the table.	T, N
13	§5	Whether the shared-noise sample is as good is "untested".	It has an answer: copying one frame's noise to F frames is not distribution-preserving. Static content is driven by noise $\sqrt{F}$ too strong (Theorem 2).	A, N
14	§5	Static map from a 2-step draft.	The draft runs on independent noise, the final sample on tied noise 11–15 dB away. The map describes a video that is never generated.	T
15	§5	41.5–43.8 dB fidelity.	Finer than the reference itself: 20 and 40 steps differ by 39.2 dB. Above about 39 dB the metric measures agreement with a discretisation error.	T
16	Table 2	Projected speedups 3.7–5.7×.	Include a fixed draft cost of 2 of 20 full steps. At the 90 % setting the draft is 56 % of the projected cost, and it caps any speedup at 10×. (Our FLOP model reproduces the projections: 3.6× and 5.6×.)	A
17	§5	Static tokens settle earlier, taken as support for the temperature view.	Predicted by pooling alone (Eq. 2), with no role for information density.	A

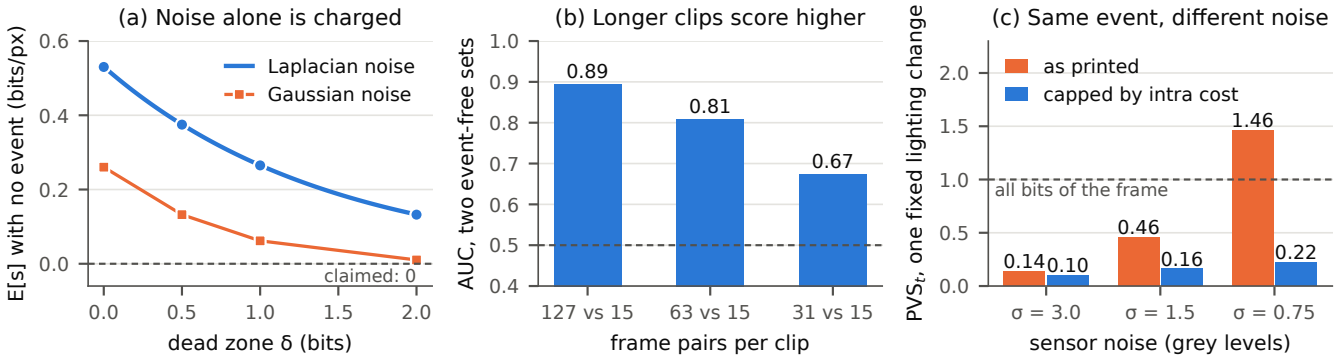


Figure 1: Three of the errata, checked numerically. (a) Expected value of Eq. 3 when the residual is pure noise; the base paper states it is zero. (b) AUC between two sets of clips that contain no event and differ only in length (dashed: no bias). (c) PVS_t of one fixed legal lighting change under three noise levels, as printed and with the per-pixel charge capped by the intra cost.

Averaging over frames gives $\bar{u}_p = (\bar{x}_p - \hat{b}_p)/t$, and subtracting gives $u[f,p] - \bar{u}_p = \delta_{f,p}/t$. Hence $\delta^{i+1} = \delta^i\,(1 - \frac{1}{Kt_i}) = \delta^i\, t_{i+1}/t_i$. The update of x_R involves x only through D. □

Theorem 1 (Averaging preserves the sample)**.** *Under the assumption of Lemma 1,* $\Phi(\textsc{mean}(\varepsilon)) = \Phi(\varepsilon)$ *for every* ε*. More generally, applying* MEAN *to the state at any step of the sampler leaves the final sample unchanged.*

Proof. MEAN keeps $(\bar{x}_S, x_R)$ and sets $\delta = 0$. By Lemma 2 the future of $(\bar{x}_S, x_R)$ is unaffected, and the output on S is $\bar{x}_S^K + \delta^K = \bar{x}_S^K$ in both cases. □

Theorem 2 (Copying heats the static content by $\sqrt{F}$)**.** *With independent noise the closed subsystem of Lemma 2 starts from* $\bar{x}_S^0 = \bar{\varepsilon}_S \sim \mathcal{N}(0, I/F)$*. After* COPY *it starts from* $\varepsilon[1,S] \sim \mathcal{N}(0,I)$*: the same sampler, but with the starting noise of the static content* $\sqrt{F}$ *times too strong. In particular, let the clip be a static Gaussian field* $x_0[f] = \sum_k c_k \psi_k$ *with orthogonal* ψ_k *and independent* $c_k \sim \mathcal{N}(0, \sigma_k^2)$*, and tie the noise on a set* S *that holds a share* ρ_k *of the energy of* ψ_k*. Then the variance of every generated coefficient is multiplied by*

$$\rho_k F + (1 - \rho_k) \quad \textit{(COPY)}, \qquad 1 \quad \textit{(MEAN)}. \tag{1}$$

Proof. The first part is the definition of the two operators. For Gaussian data D is linear, hence so is Φ, and the generated coefficient is $c_k = g_k \langle \varepsilon, u_k \rangle$ with u_k the unit vector along ψ_k repeated over frames and g_k a gain that does not depend on ε. For independent noise $\mathrm{Var}\langle \varepsilon, u_k \rangle = 1$. After COPY the F frames contribute coherently on S, which gives $\rho_k F^2/F + (1 - \rho_k)$; after MEAN the projection is unchanged. □

Reading. Proportional attention tells the network that one key stands for F tokens. A network trained on independent noise has learned to pool F tokens as F independent measurements of the same background, which is correct, and gives the pooled value a noise level of $t/\sqrt{F}$. COPY hands it a single measurement counted F times: the network is over-confident by $\sqrt{F}$ and treats noise as signal. MEAN hands it the pooled measurement itself. The change to the method is one line, and the compute saving is untouched because the copies are identical either way.

Static tokens are colder. In the closed subsystem a static token is $\bar{x} = (1-t)\,b + t\,\bar{\varepsilon}$ with $\mathrm{Var}(\bar{\varepsilon}) = 1/F$. At time t it has the signal-to-noise ratio of an ordinary token at the earlier-looking time

$$t' = \frac{t}{t + \sqrt{F}\,(1-t)} < t. \tag{2}$$

This makes the base paper's temperature picture quantitative, and it predicts its settling measurement without any appeal to information density: if moving tokens settle at step 7 of 20 ($t = 0.65$), static tokens of an 8-frame clip reach the same signal-to-noise ratio at $t = 0.84$, i.e. after about 3 steps; the base paper measured 4.

Collapsing mid-generation. The second sentence of Theorem 1 removes the need for a separate draft pass. One can run the first k steps of the real trajectory on all tokens, read the static map from the clean prediction the sampler has just computed, replace the static tokens by their frame mean, and continue with sharing. Those k steps are not overhead, because they are the trajectory. The base paper reported that switching mid-generation fails because "switching resets each copy's trajectory"; that is the signature of COPY, which replaces the pooled state by one frame's state.

When clips have different static sets. The assumption of Lemma 1 fixes S. Real data mix clips with different static sets, and then the deviations δ carry evidence about *whether* S is static. Take a clip a that is static on S and a clip c that is not, with deviations δ^c on S. Repeating the computation of Lemma 1, their posterior log-odds given x are

$$\log \frac{w_c}{w_a} = G(\bar{x}_S, x_R) - \frac{(1-t)^2 \|\delta^c\|^2}{2t^2} + \frac{1-t}{t^2}\,\langle \delta, \delta^c \rangle. \tag{3}$$

If x is a noised version of a clip static on S, then $\delta = t\,n$ with n pure noise and the last term is $\mathcal{N}\big(0, (1-t)^2 \|\delta^c\|^2/t^2\big)$: a zero-mean fluctuation in the evidence against staticness. MEAN sets this term to its mean and leaves the weights among

clips static on S untouched. So averaging is no longer exact, but its only effect is to remove a zero-mean term; the effect vanishes once the posterior has committed, which is why a later collapse is safer (§5). COPY removes the same term and, in addition, rescales the noise of $\bar{x}_S$ by $\sqrt{F}$.

4 Mean-tied frame sharing

The method keeps every part of the base paper's frame sharing and changes how the copies are made identical.

1. **Map.** Decide which positions are static. Either (a) run a 2-step draft from the *same* noise and threshold its largest frame-to-frame change, as in the base paper; or (b) run the first k steps of the real trajectory on all tokens and threshold the change of the clean prediction $\hat{x}_0 = x - t\,v$ that step $k-1$ produced anyway.
2. **Collapse.** At every static position replace the F copies by their frame mean: of the starting noise for (a), of the current state for (b). The base paper uses the leader frame's value here.
3. **Share.** Keep one token per static position, add $\log F$ to its attention logit as a key, and write its output to all F frames. Unchanged.

With variant (a) the mean-tied noise has standard deviation $1/\sqrt{F}$ at static positions. With variant (b) there is no extra pass: the k warm-up steps are part of the trajectory, and by Theorem 1 the collapse does not disturb it. Because averaging preserves the sample, the draft and the final video are now the *same* sample, so the static map describes the video that is actually produced (Table 1, item 14), and the accelerated output can be scored against the original output like any other acceleration method (item 12).

What the theory does not cover. The guarantees assume an ideal denoiser and a truly static S. A trained network is ideal only near its training distribution, and a state with $\delta = 0$ is outside it: each tied token is cleaner than the time t announces. Token removal with proportional attention adds its own error (item 11). A wrongly marked position is forced to be static. The experiments measure all three, and the first turns out to matter (§6).

A sufficiency regulariser. Lemma 1 is a property of the ideal denoiser that a finite network has no reason to learn exactly, because independent noise never presents it with $\delta = 0$. It can be taught. During training we show every clip twice with the same noise ε: once untouched and once with the noise on a random subset S of the clip's static positions replaced by its frame mean, $\tilde{\varepsilon} = \text{MEAN}_S(\varepsilon)$. With v and $\tilde{v}$ the network's velocities on the two views, the loss is

$$\tfrac{1}{2}\|v-(\varepsilon-x_0)\|^2 + \tfrac{1}{2}\|\tilde{v}-(\tilde{\varepsilon}-x_0)\|^2 + \lambda\,\|\text{MEAN}_S(v)-\tilde{v}\|^2. \tag{4}$$

The first two terms are the ordinary rectified-flow loss on each view. The third is zero for the ideal denoiser: by Lemma 2 the frame mean of the velocity on S is the velocity of the pooled token, and elsewhere the two velocities coincide. It asks the network to treat the deviations as the nuisance they are.

5 Checks that need no network

Exact Gaussian model. We take the six-mode Gaussian background field of our data set (§6) as the whole clip, repeat it over $F \in \{2,4,8,16\}$ frames, and tie the starting noise on the top 100, 75 or 50 % of the rows. The ideal denoiser is linear and known in closed form, so the 20-step sampler is exact. Over all 72 coefficient–configuration pairs the measured variance inflation under COPY matches Eq. 1 within 4.2 % (sampling error of 20,000 clips); for eight fully tied frames it is 7.98 against a predicted 8 (Fig. 2a). Under MEAN the inflation is 1 and the output equals the independent-noise output to 1.8×10^{-15}.

Exact nonlinear denoiser. For a finite data set the ideal denoiser is a softmax-weighted average of the data [3], and the sampler lands on one of the clips. We use 4096 clips from our generator (smooth background, one or two moving sprites, a different static set in every clip, so this is the mixture case of Eq. 3) and 1024 noise seeds. For each seed we record which clip independent noise produces, tie the noise on that clip's static set, and ask whether the sampler still produces *that* clip (Table 2).

With COPY it does so in 0.1 % of runs, at 10.4 dB from the original, which is the 11–15 dB the base paper reports; the backgrounds of the clips it lands on are 1.73× as dispersed as they should be. With MEAN it produces the original clip in 90.2 % of runs with a dispersion ratio of 1.01. The remaining runs are the mixture effect of Eq. 3; collapsing a few steps later removes it (100.0 % at step 6, against 84.1 % for COPY). A control separates the two things MEAN does. Scaling the leader's noise by $1/\sqrt{F}$ gives the right *variance* but not the right *value*: the dispersion is correct (0.98) and the sample is a different one (12.3 % same clip). The variance fixes the distribution; taking the actual mean keeps the sample.

The 2-step draft is a poor map here. An ideal denoiser at $t = 0.5$ returns a blend of many clips, and no threshold reaches more than 68 % precision; forcing a moving position to be static changes the sample under either operator. Reading the map from the trajectory a few steps later is far more reliable, and Theorem 1 makes that free.

6 A trained transformer

Setup. We follow the base paper's protocol as closely as its text allows. Clips have 8 frames of 32×32 grayscale pixels and are cut into 8×8 patches, giving 128 tokens. The background is a static Gaussian random field in six smooth modes with known variances, so that its spread in generated clips can be measured; one or two textured sprites move over it with sub-pixel motion. On average 62 % of the 16 positions are static. The model is a diffusion transformer [4] with 1.27M

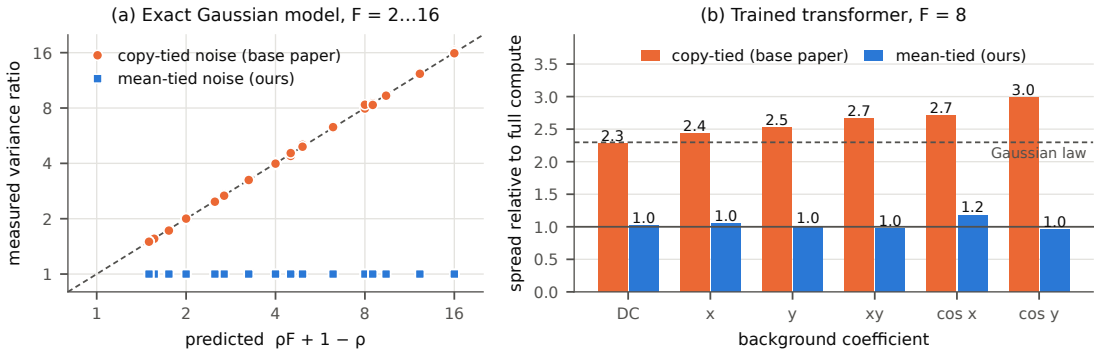


Figure 2: The variance law. (a) Exact Gaussian model: measured against predicted inflation of every background coefficient for $F \in \{2,4,8,16\}$ and three tied shares; mean-tied noise sits at 1. (b) Trained standard model, oracle map: spread of the generated background coefficients relative to full compute.

Table 2: Exact nonlinear denoiser on 4096 clips, 1024 seeds, no network. "Same clip": the sampler lands on the clip that independent noise would have produced. "Spread": dispersion of the background coefficients of the produced clips relative to independent noise (1 is correct).

			COPY (base)		MEAN (ours)	
Static map	tied at step	tied (%)	same clip (%)	spread	same clip (%)	spread
oracle	0	63	0.1	1.73	**90.2**	1.01
2-step draft	0	60	1.3	1.73	**45.2**	0.99
trajectory	3	18	33.2	1.10	**98.1**	1.00
trajectory	4	41	38.1	1.07	**99.1**	1.00
trajectory	6	64	84.1	1.01	**100.0**	1.00

parameters (4 layers, width 128, adaLN-zero conditioning, additive spatial and temporal position embeddings), trained with rectified flow [5] for 12k steps on a CPU and sampled with 20 Euler steps. Speedups are analytic FLOP counts that include every draft and warm-up call; "projected" applies the same token counts to the base paper's 75.6k-token, 40-layer, width-3072 model. Token removal is executed through an attention bias, which a unit test shows to be identical to physical removal. Fidelity is the median PSNR (peak-to-peak 2) over 1,024 noise seeds.

Every method is scored against the *original* sample: 20-step full compute from the untouched noise. We also report the base paper's own metric, the spread of the generated background coefficients relative to full compute, and the texture (mean absolute difference from a 3×3 blur) and motion (mean absolute frame difference) of the generated clips relative to full compute. Static maps come from an oracle (the original sample itself), from a 2-step draft, or from the trajectory at the switch step; thresholds are set on separate seeds, to 95% precision against the oracle, or to the best F1 ("aggressive"). As in the base paper, the model's own samples are imperfect; they are under-dispersed (background spread 0.70 of the data's) and show patch seams.

6.1 Standard training: the base paper's result, on both footings

Table 3 (left) and Fig. 3a give the picture.

Copy-tied sharing reproduces the base paper. Against full compute from its own tied noise, copy-tied sharing scores 61.2 dB; against the original sample it scores 7.9 dB. These are the two numbers of the base paper (43.6 and 11–15 dB), and only the first was put in its comparison. The copied noise does what Theorem 2 says: the background is 2.6× too dispersed, where the Gaussian law predicts 2.3 for the tied share (Fig. 2b), and the static patches fill with high-contrast texture, 5.2× the normal amount (Fig. 4). The network has been handed noise and told it is signal.

Mean-tied sharing removes the over-dispersion. With the same map, the same token removal and the same FLOPs, averaging brings the background spread to 1.03 and the fidelity to the original from 7.9 to 20.9 dB. The control that scales the leader's noise by $1/\sqrt{F}$ again gives the right spread (0.98) and a different sample (15.6 dB).

But it is not sample-preserving, and it loses to fewer steps. 20.9 dB is far from exact. At the same speedup, simply taking fewer steps is closer to the original sample (26.9 dB at 2×, 23.2 dB at 2.9×), and this holds for every map we tried (Fig. 3a). The loss is not in the compute reuse: full compute from mean-tied noise, with no token removed, lands at the same 20.9 dB, and sharing reproduces *that* to 66.8 dB. It is in the network's response to tied tokens. Mean-tied samples carry 2.4× the texture and 2.5× the motion of full-compute samples: the error has moved from the tied tokens into the untied ones. So on a network trained in the standard way, the base paper's claim that frame sharing beats fewer steps does not survive an equal footing, with either way of tying the noise.

6.2 Why: the network is not invariant to the deviations

Lemma 1 says the ideal denoiser ignores the deviations δ. We test this directly by scaling the deviations of the static starting

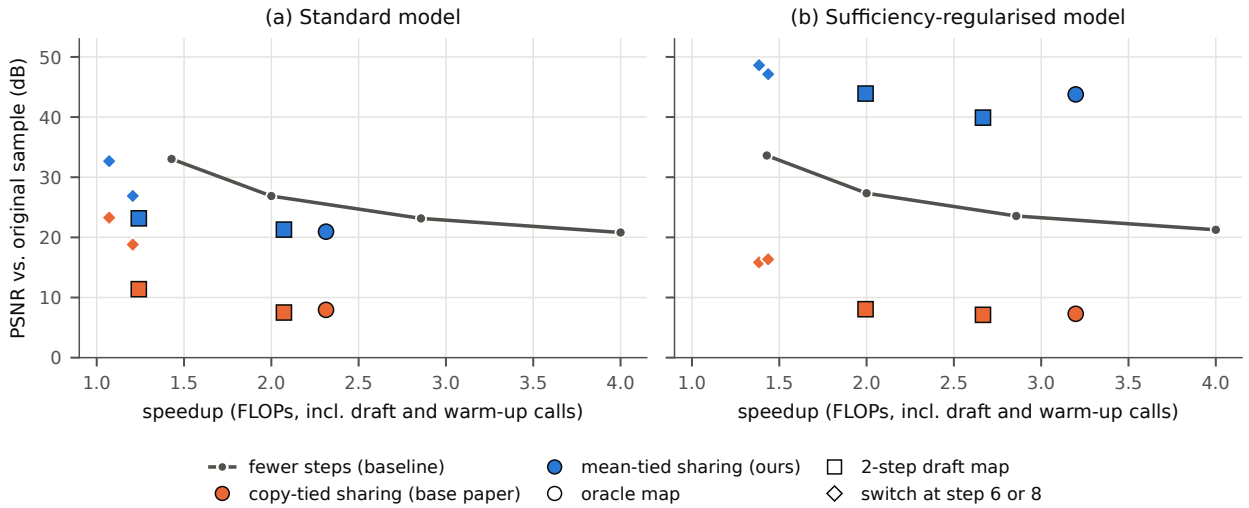


Figure 3: Fidelity to the original sample against speedup, the base paper's Fig. 4 redrawn with every method scored against the same reference. (a) Standard training: both ways of tying lie below the fewer-steps baseline; averaging is about 13 dB above copying. (b) After the sufficiency fine-tune: mean-tied sharing lies 15–20 dB above the baseline; copying does not move.

Table 3: Frame sharing on equal footing with the baseline. PSNR is against the original 20-step full-compute sample from the untouched noise (median over 1,024 seeds). "bg", "tex." and "mot.": background spread, texture and motion relative to full compute (1 is correct). Left: standard training. Right: the same network after a short fine-tune with the sufficiency regulariser (Eq. 4).

	Standard model						Sufficiency-regularised model					
Method	shared (%)	speed-up	proj.	PSNR (dB)	bg	tex. / mot.	shared (%)	speed-up	proj.	PSNR (dB)	bg	tex. / mot.
Baseline												
Fewer steps, 20→14	–	1.43	1.4	33.0	0.88	0.9 / 0.9	–	1.43	1.4	33.6	0.88	0.9 / 0.9
Fewer steps, 20→10	–	2.00	2.0	26.9	0.75	0.8 / 0.8	–	2.00	2.0	27.3	0.74	0.9 / 0.8
Fewer steps, 20→7	–	2.86	2.9	23.2	0.59	0.7 / 0.7	–	2.86	2.9	23.6	0.59	0.8 / 0.8
Fewer steps, 20→5	–	4.00	4.0	20.8	0.46	0.6 / 0.6	–	4.00	4.0	21.3	0.46	0.8 / 0.8
Frame sharing, copy-tied noise (base paper)												
oracle map	61	2.31	3.6	7.9	2.60	5.2 / 1.0	75	3.20	5.9	7.3	2.77	8.1 / 1.0
2-step draft map	31	1.24	1.5	11.4	2.05	3.1 / 1.1	65	1.99	2.9	8.1	2.63	7.1 / 1.1
2-step draft map, aggressive	67	2.07	3.0	7.5	2.60	5.7 / 1.0	80	2.67	4.2	7.1	2.79	8.6 / 0.9
switch at step 8, no draft	30	1.21	1.3	18.8	1.08	2.4 / 0.9	54	1.44	1.7	16.4	1.09	4.8 / 0.7
Frame sharing, mean-tied noise (ours)												
oracle map	61	2.31	3.6	**20.9**	1.03	2.4 / 2.5	75	3.20	5.9	**43.8**	1.00	1.0 / 0.9
2-step draft map	31	1.24	1.5	**23.2**	1.07	2.0 / 2.2	65	1.99	2.9	**43.9**	1.00	1.0 / 0.9
2-step draft map, aggressive	67	2.07	3.0	**21.3**	1.01	2.2 / 2.2	80	2.67	4.2	**39.9**	0.99	1.0 / 0.8
switch at step 8, no draft	30	1.21	1.3	**26.9**	1.04	1.7 / 1.8	54	1.44	1.7	**47.1**	1.00	1.0 / 0.9

noise by a factor α and generating with full compute; an ideal denoiser returns the same sample for every α. The trained network does not (Table 4): shrinking the deviations by only 10 % already moves the sample to 32.8 dB, and removing them gives 20.9 dB. Its *consistency error*, the mean squared difference between the frame-mean velocity on an untouched input and the velocity on the mean-tied input, is 8.8×10^{-3} where the ideal value is zero. Longer standard training barely changes this (19.7 dB at 6k steps, 20.9 dB at 12k, 21.3 dB after 3k more). A network that has only ever seen independent noise uses the deviations as a cue, most plausibly for how noisy its input is; a tied token looks too clean for its time step.

6.3 Teaching sufficiency

We fine-tune the standard model for 3k steps (a quarter of its training) with the regulariser of Eq. 4, $\lambda = 50$. The consistency error falls by a factor of 73 (Table 4), and the right half of Table 3 and Fig. 3b change character.

Mean-tied sharing becomes sample-preserving. With the oracle map it shares 75 % of the positions, uses 3.2× fewer FLOPs (5.9× projected), and reproduces the original sample at 43.8 dB (10th percentile over seeds 40.1 dB). With the 2-step draft map of the base paper, which needs no oracle, it reaches 43.9 dB at 2.0× and, at the aggressive threshold,

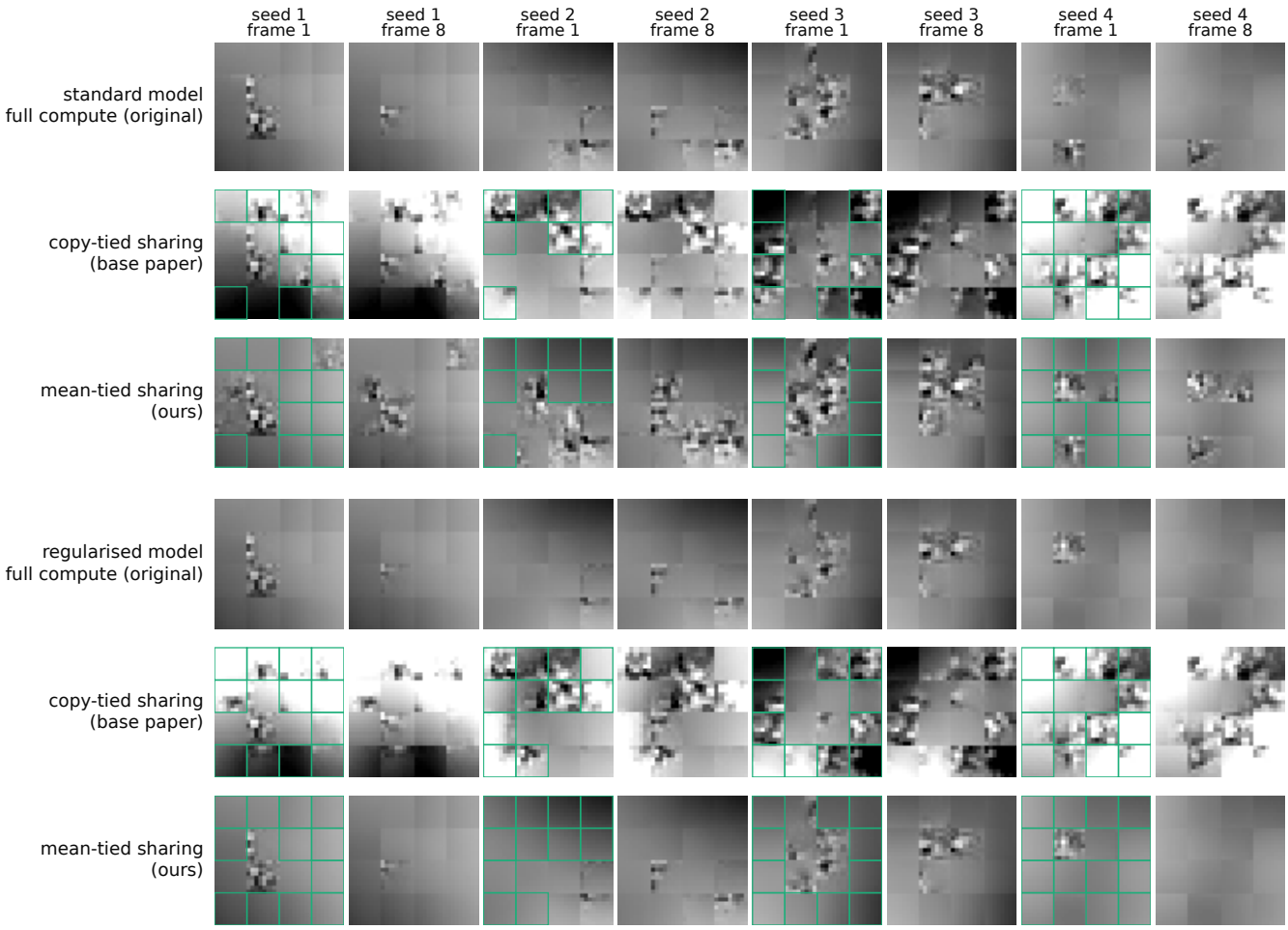


Figure 4: Four seeds, first and last frame, oracle map (shared positions outlined in the first frame). Copy-tied sharing fills the shared patches with saturated texture on either model: noise taken for signal. Mean-tied sharing gives plausible but different clips on the standard model and reproduces the original clips on the regularised one. The regularised model's sprites are fainter, the cost discussed in §6.

39.9 dB at 2.7× (2.9× and 4.2× projected), draft included; the fewer-steps baseline is at 27.3 dB at 2× and 23.6 dB at 2.9×. Texture is unchanged (1.01 of full compute) and so is motion on the positions that are not shared (1.05); the overall motion measure is 0.94 because full compute leaves a faint frame-to-frame jitter on static positions, 10 % of its motion, which sharing removes by construction. Switching mid-generation, which the base paper reports as a failure, now works without any draft: collapsing to the mean at step 8 gives 47.1 dB, where collapsing to the leader gives 16.4 dB. These are the fidelities the base paper reported for frame sharing, measured this time against the sample the model would have produced anyway.

Copying is not rescued. On the same network, copy-tied sharing stays at 7.3 dB with backgrounds 2.8× too dispersed. The regulariser makes the network a better approximation of the ideal denoiser; it cannot make a single noise sample look like the mean of eight.

Proportional attention matters, and is not exact. Without the $\log F$ bias, mean-tied sharing falls to 19.6 dB. With it, one forward call on tied tokens differs from full attention by 0.15 % in the standard model against 18 % without it: a large gain, as the base paper found, and not an identity (Table 1, item 11).

6.4 What the regulariser costs

The regulariser changes the generator. Table 4 compares full-compute samples of each model variant with the training data. Continuing standard training for the same 3k steps changes nothing: the consistency error stays at 8.7×10^{-3}, mean-tying at 21.3 dB, texture and motion as before. The effect of the fine-tune is the regulariser's, not that of more training.

Its price is texture. With $\lambda = 50$, full-compute samples carry 0.54 of the data's texture where the standard model carries 0.85, and the regression loss on untouched inputs is 2.0 % higher than the control's. The sprites are fainter (Fig. 4); more positions are static, which is part of why the regularised model shares more. In motion the two models err in opposite directions: the standard one moves too much (1.46 of the data's frame difference) and the regularised one too little (0.71). Background spread is unchanged.

Table 4: Model variants. "Cons.": consistency error $\times 10^3$ (0 for an ideal denoiser). "α=0.9" and "mean": PSNR against the original sample after scaling the frame deviations of the static starting noise by 0.9 and by 0 (full compute, oracle map); "copy": after copying the leader's noise. Last three columns: texture, motion and background spread of full-compute samples relative to the training data.

		PSNR vs. original (dB)			vs. data		
Model	Cons.	α=0.9	mean	copy	tex.	mot.	bg
Standard, 3k	13.5	35.6	23.3	11.2	1.48	2.11	0.76
Standard, 6k	11.8	29.5	19.7	8.0	0.93	1.30	0.65
Standard, 12k	8.8	32.8	20.9	7.9	0.85	1.46	0.67
+3k, λ=0	8.7	33.4	21.3	7.9	0.85	1.47	0.68
+3k, λ=5	0.5	42.3	37.1	7.9	0.74	1.24	0.67
+3k, λ=50	0.1	48.0	43.5	7.1	0.54	0.71	0.65

A gentler setting keeps most of the benefit. With $\lambda = 5$ the consistency error still falls 19-fold, mean-tying reaches 37.1 dB, the regression loss is within 0.4 % of the control's, texture drops less (0.74) and motion moves towards the data (1.24). On that model, mean-tied sharing with the 2-step draft map reproduces the original sample at 39.4 dB with 1.4× fewer FLOPs (35.2 dB at 2.1× with the aggressive threshold; fewer steps: 27.0 and 23.3 dB at 2× and 2.9×), and copying stays at 8.0 dB. We tried only these two weights. We suspect the network buys part of its invariance by leaning less on its noisy input, which is the cheapest way to ignore the deviations and also blurs the sprites; the ideal denoiser shows that this is not necessary, so longer training or a better-placed weight may remove the cost. That is open.

6.5 Distribution test

Per-sample fidelity cannot say whether an accelerated sampler still draws from the model's distribution when it does not reproduce the sample. We train a small 3D CNN to tell 2,048 accelerated samples from 2,048 full-compute samples on fresh seeds (classifier two-sample test [6]; 0.5 means indistinguishable; white noise of standard deviation 0.03 is added to both sides so that the faint jitter full compute leaves on static positions is not a cue). On the standard model (Table 5), copy-tied sharing is the most detectable sampler we tested (0.96) although it saves the least. Mean-tied sharing is less detectable (0.86) but no better than 7-step sampling (0.85): the texture excess of Table 3 is a real shift of the distribution. On the regularised model the classifier cannot tell mean-tied sharing from full compute (0.50, control 0.51), while copy-tied sharing (1.00) and fewer steps (0.71, 0.84) remain detectable.

6.6 Two further checks

Static tokens settle earlier, roughly as predicted. In full-compute runs of the standard model, the clean prediction of a moving token comes within 0.05 RMS of its final value at step 8 of 20 (median), that of a static token at step 3. Eq. 2 maps step 8 to step 3.8 for $F = 8$. The base paper measured

Table 5: Classifier two-sample test against 20-step full compute of the same model (held-out accuracy, mean of 3 classifier seeds; 0.5 is indistinguishable).

Sampler	Standard	Regularised
Full compute, fresh seeds (control)	0.51	0.51
Fewer steps, 20→10	0.72	0.71
Fewer steps, 20→7	0.85	0.84
Copy-tied sharing, 2-step draft	0.96	1.00
Mean-tied sharing, 2-step draft	0.86	0.50

7 and 4. Static content is also simpler, so this is consistent with the clock rather than a test of it.

FLOPs and time. The FLOP model is not a measurement, so we timed the standard model on a CPU with the shortened sequences built physically (batch 32, 4 threads). With 8, 12 and 16 of 16 positions shared, the forward pass is 2.4×, 3.8× and 8.1× faster, where the FLOP model says 1.9×, 3.2× and 9.1×. So at this size the count is a fair guide. It says little about a large model on a GPU, where attention dominates and memory traffic matters.

7 Related work

Token merging and caching. ToMe [2, 7] merges similar tokens and introduced proportional attention; VidToMe [8] merges across frames for editing. Caching methods reuse features across sampling steps [9, 10]. The base paper [1] removes tokens before generation, which requires identical copies and hence tied noise. Our contribution is orthogonal to the choice of merge rule: it concerns the noise.

Correlated noise for video. PYoCo [11] trains with a noise prior that shares a component across frames; because the model is fine-tuned on that prior, its denoiser is calibrated to it. FreeNoise [12] reschedules noise across windows at inference. Noise warping [13, 14] transports noise along motion while keeping every frame's noise white, since the per-frame marginal is what the model was trained on. Frame sharing is a different regime: the network is told, through proportional attention, that a token stands for F tokens, and the consistent value for it is the mean of F white samples, with variance $1/F$. Tying at unit variance, which is correct per frame, is what makes the copies too hot.

Per-token noise levels. Diffusion Forcing [15] trains with independent noise levels per token. Eq. 2 says a merged static token is exactly such a token, running on a faster clock. A model trained to accept one token per static position at its own time t' would see in-distribution inputs, which mean-tying on a standard model does not provide; we leave this to future work.

Ideal denoisers. The closed-form posterior mean over a finite data set [3] gives a network-free test bed, which we use in §5. The sufficiency argument behind Lemma 1 is Rao–Blackwellisation: the deviations are ancillary, and discarding

them loses nothing.

8 Limitations

Everything here is toy-scale: a 1.27M-parameter transformer, $8 \times 32 \times 32$ grayscale synthetic clips, a static camera, CPU only. We have not run an open video model, so nothing here shows that mean-tying helps at scale; the argument that it should rests on the theory, which does not depend on scale, and on the network-free checks. Each model variant was trained once, and the regulariser was tried at two weights only; it lowers the texture of the model's samples, and we do not know whether that can be avoided. The errata on the permanence score come from our re-implementation of the printed equations, not from the authors' code, and we did not rebuild their benchmark; items marked **C** in Table 1 are concerns, not findings. We propose no replacement score. The classifier two-sample test is a lower bound on distinguishability. Speedups are analytic FLOP counts; the timing check in §6 covers only a small model on a CPU.

9 Conclusion

The base paper asked whether a shared-noise sample is as good as an independent-noise one. For noise shared by copying, the answer is no, for a reason that can be stated in one sentence: the frame mean of F white samples has variance $1/F$, and the network knows it. Sharing by averaging respects that, is exact for an ideal denoiser, and costs nothing. On a trained network the missing ingredient is the invariance that the ideal denoiser has for free. Once it is taught, frame sharing does what the base paper's headline claimed, this time against the right reference, and a classifier cannot tell its samples from full compute. What remains open is the price of that invariance in sample quality, the static-camera assumption, a reliable static map early in the trajectory, and above all scale.

Reproducibility. Code, trained weights, every result as JSON, and the scripts that turn them into the tables, figures and numbers of this report are in the accompanying repository; `FINDINGS.md` maps each item to its script. Numbers from our experiments are inserted from the result files by script; numbers attributed to the base paper are quoted from its text.

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